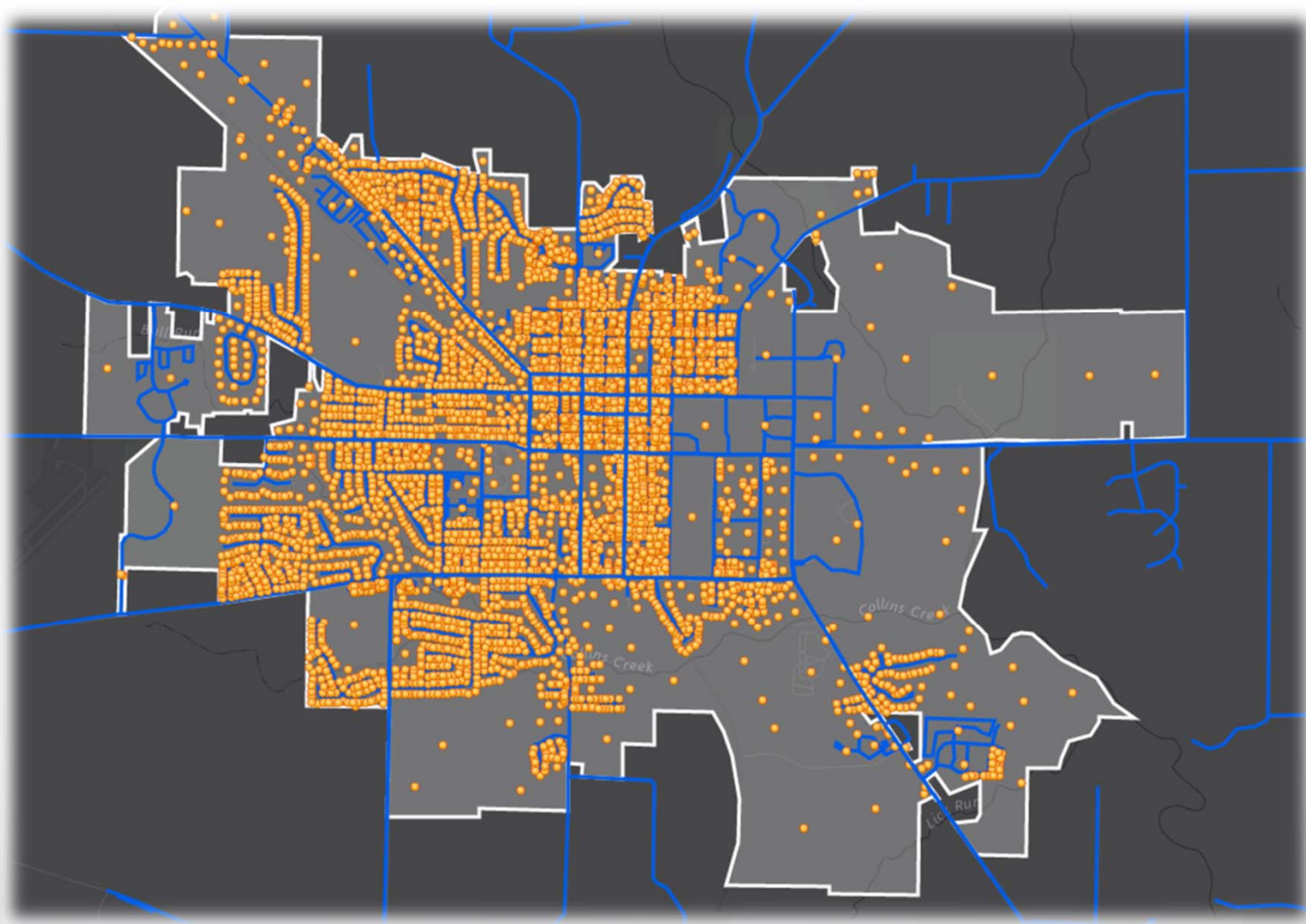


**Michael Baker**

**INTERNATIONAL**

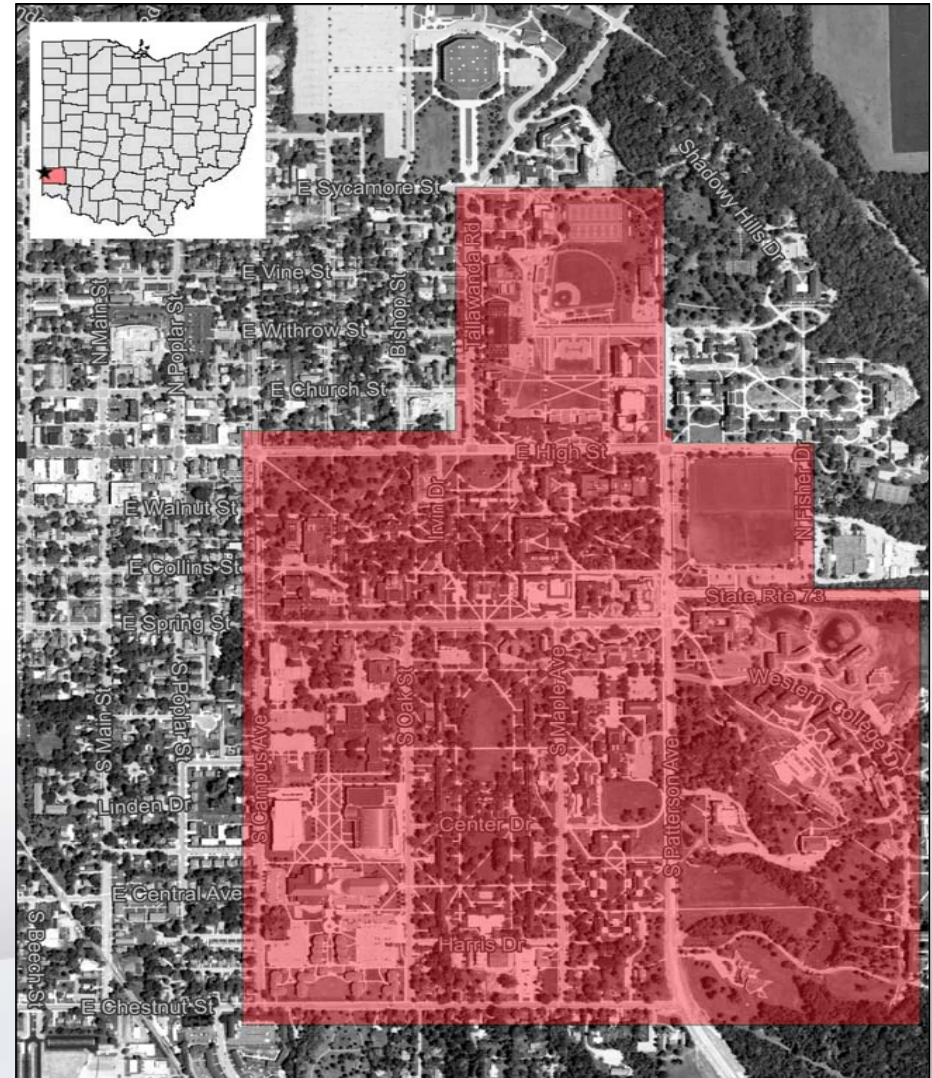
*We Make a Difference*



# Application of Streetlight Data for Miami University Circulation Study and Wisconsin DOT

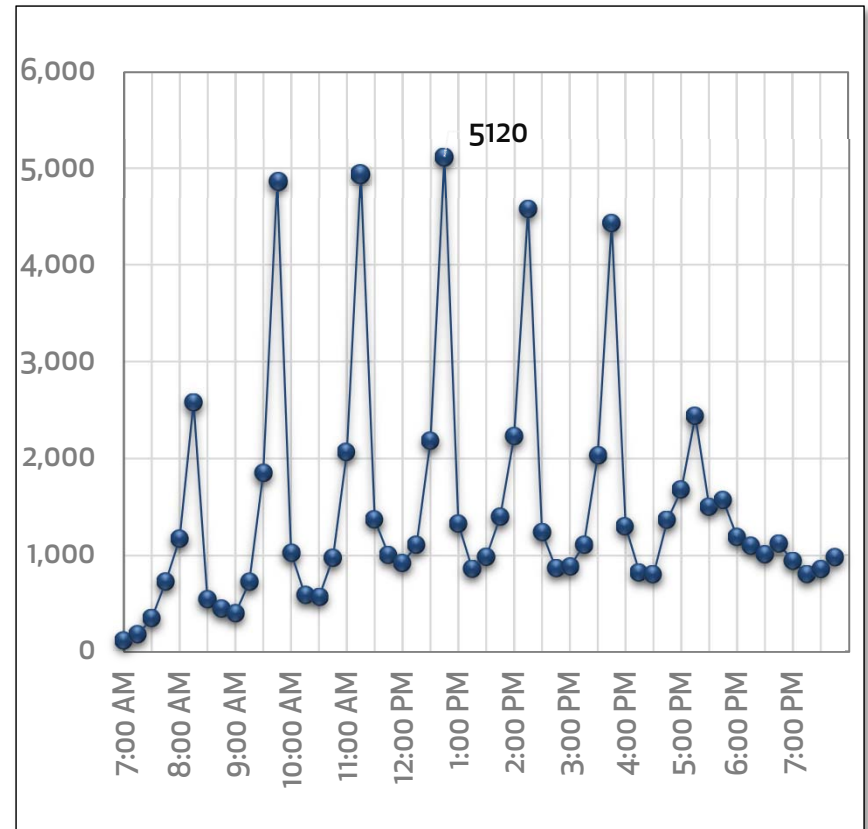
*Angela M. Coates, PE*

# Miami University Campus Circulation Study



# Project Background

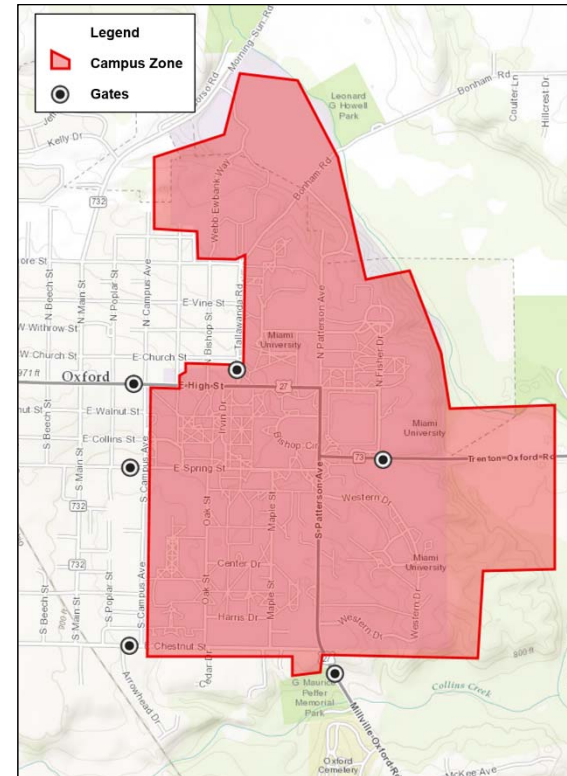
- During class change time, pedestrians/bicycles overtake the campus resulting in congestion and safety issues
- Goal: To compile a set of data and tools to inform several programmatic decisions, both current and future



15-Minute Pedestrian Crossings

# Approach to Streetlight Data

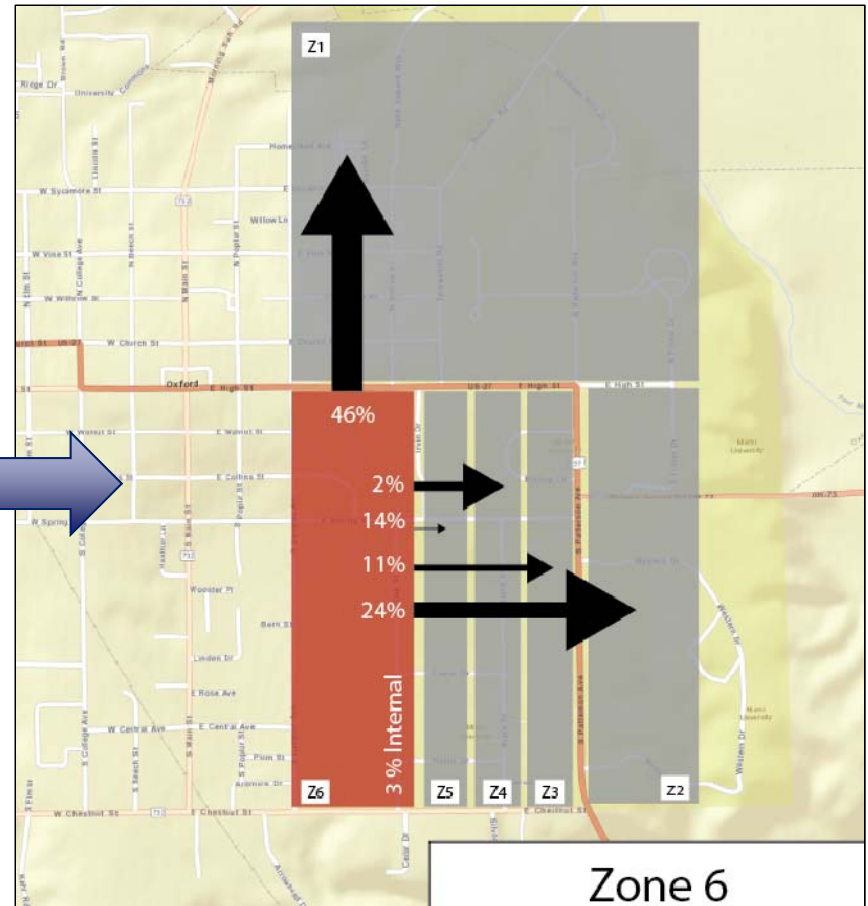
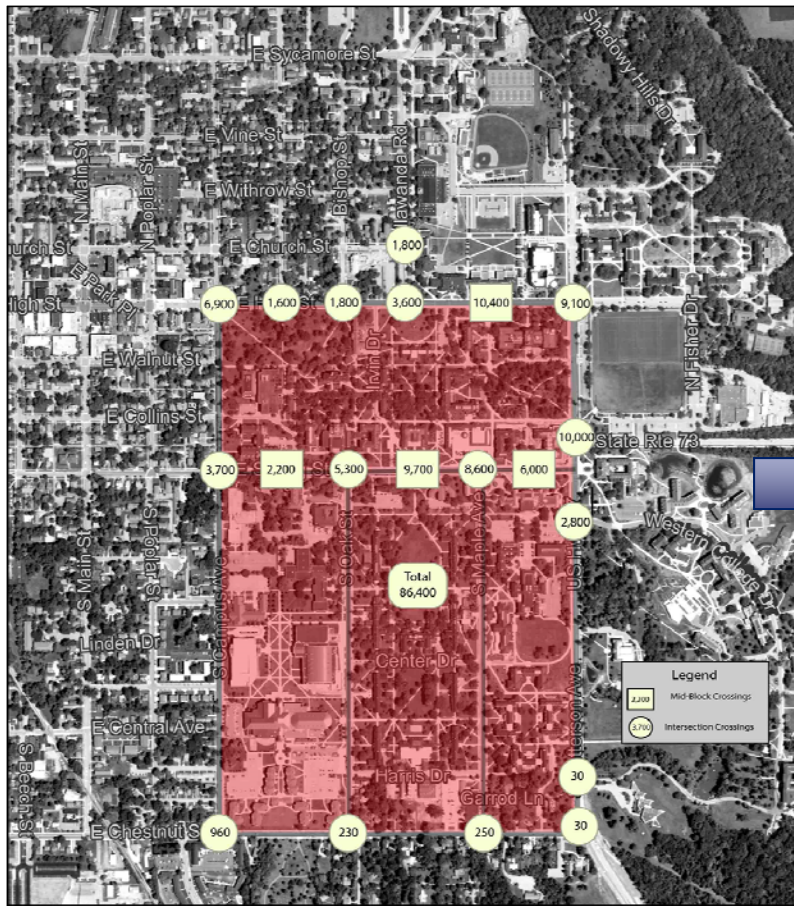
- Data Periods
  - Fall and Spring Semester Only
- Day Types
  - Average Weekday Only
- Day Parts
  - Hourly Breakdown



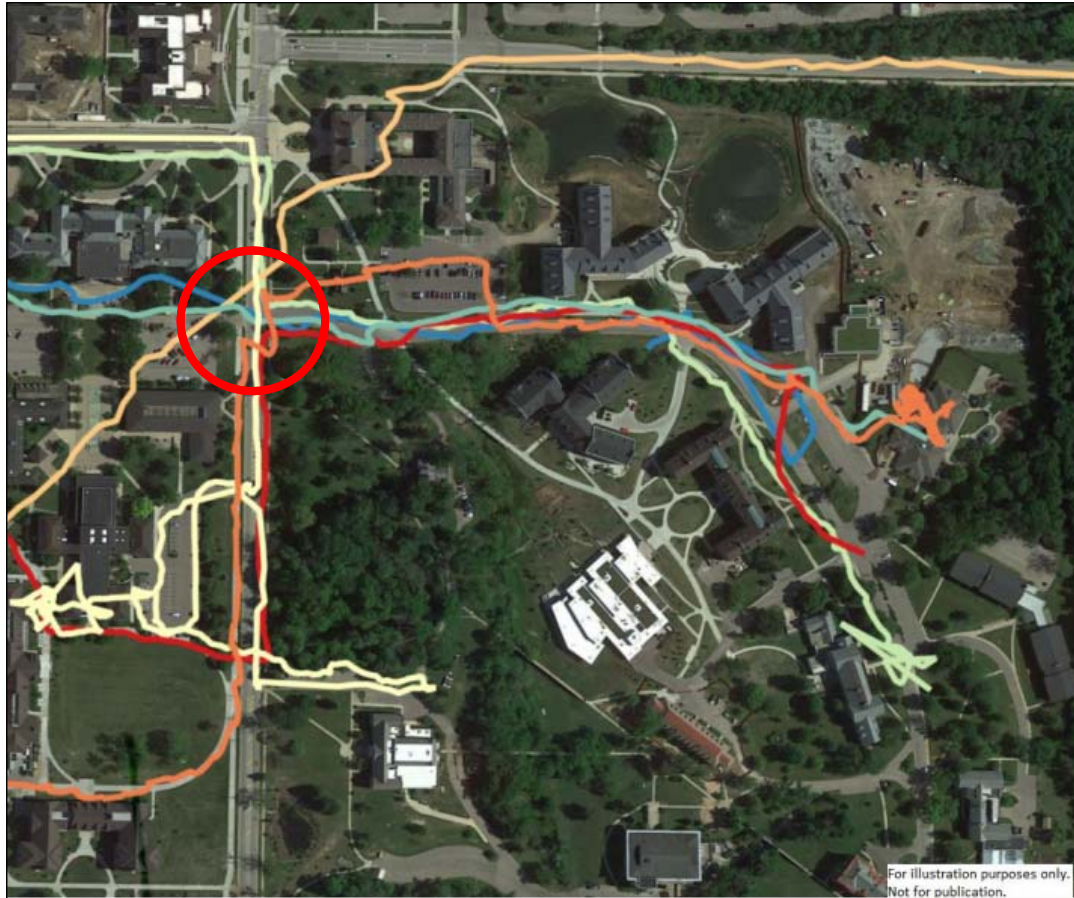
Campus Zone Set



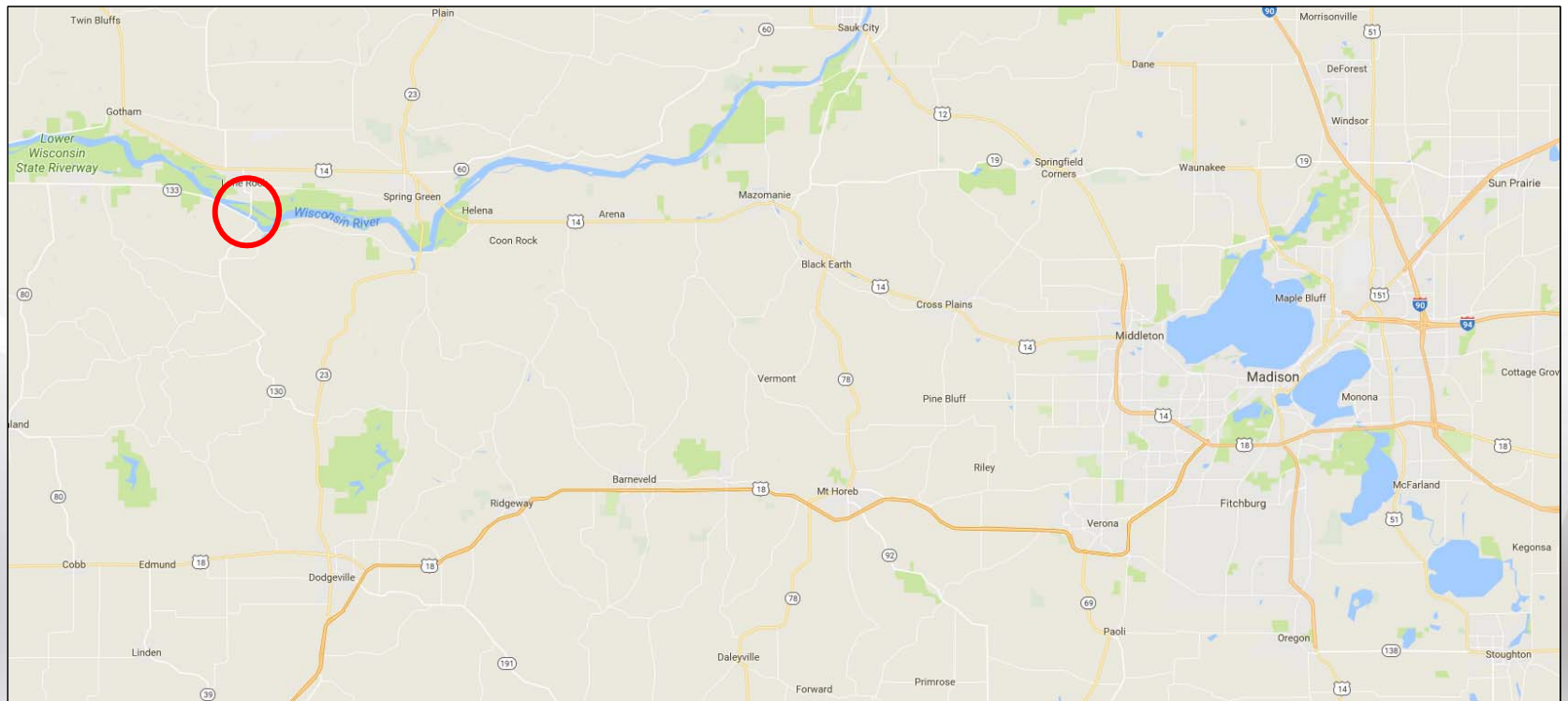
# Bike/Ped OD Patterns



# Illustrative Trip Traces



# Wisconsin River Crossing WIS-130 Lone Rock, WI





# Project Background

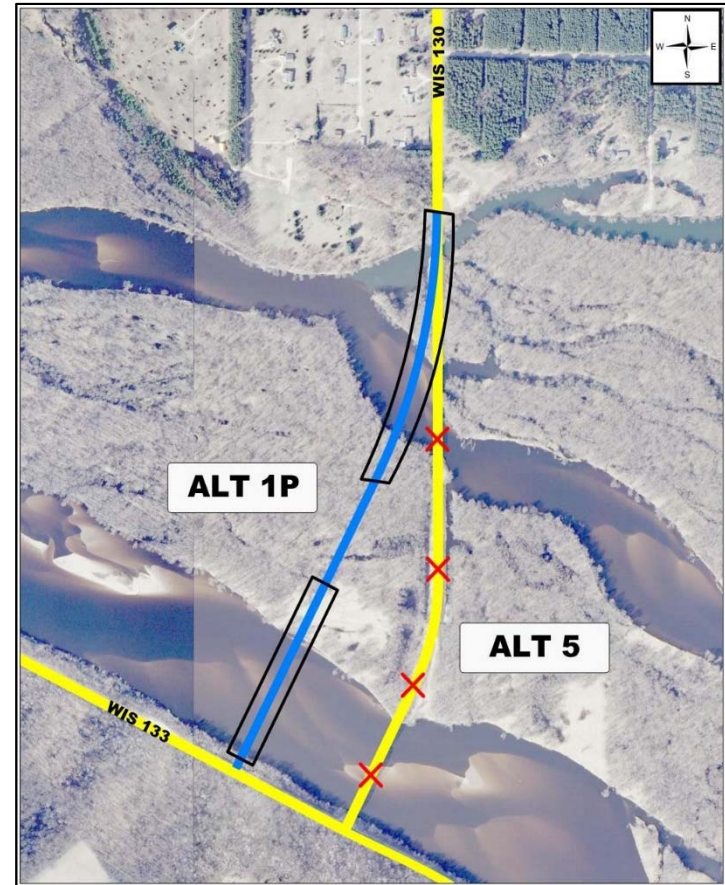


- The WIS-130 Wisconsin River crossing consists of three bridges that were built between 1932 and 1942.
- Two of the bridges are structurally deficient.
- Three are functionally obsolete with narrow clear widths ranging from 20 to 24 feet.



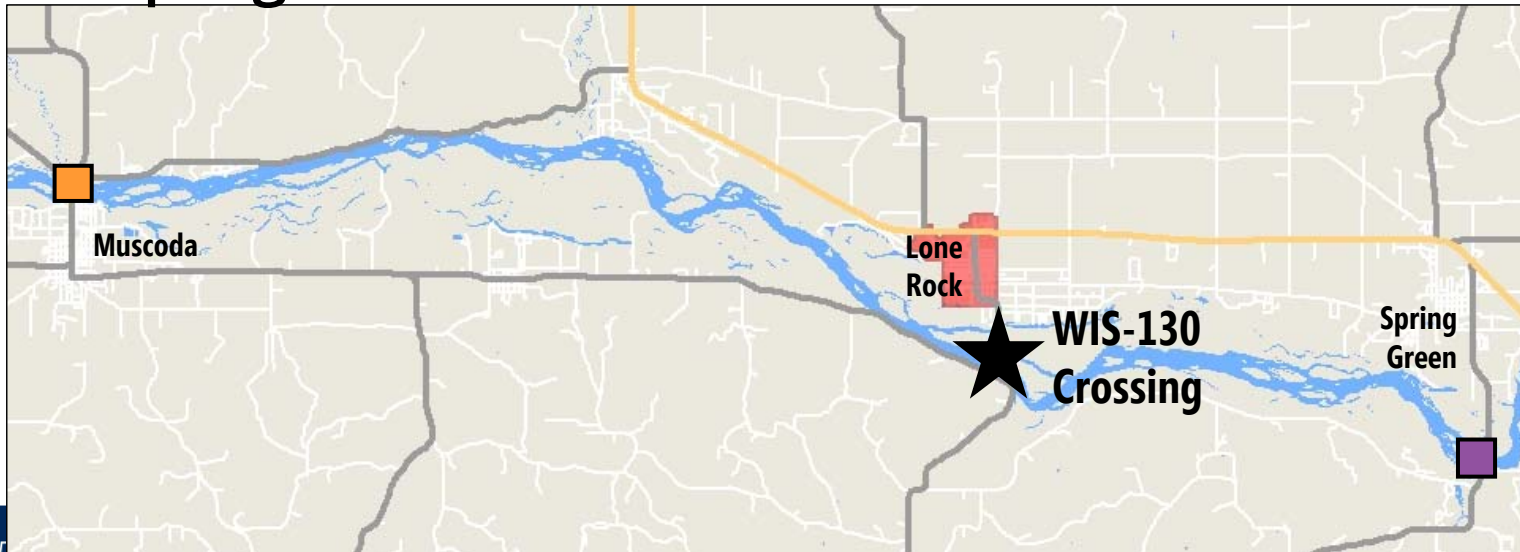
# Project Background

- Purpose of the project was to address the deteriorating structures across the Wisconsin River while avoiding/minimizing any impacts.
- Two top alternatives:
  - Alt. 1P – Near adjacent structure reconstruction with two bridges
  - Alt. 5 – Planned elimination at end of service life (2028)



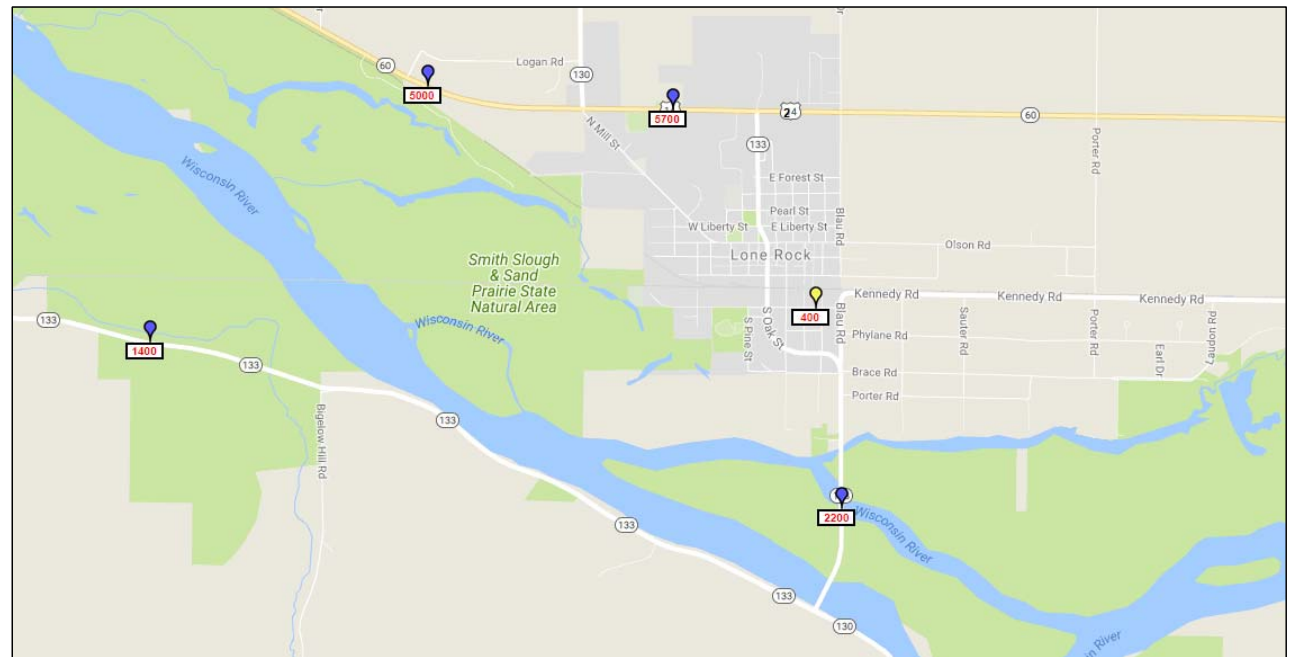
# Project Summary

- WIS-130 crossing provides direct access to/from the Village of Lone Rock, WI.
- Alternative river crossings:
  - Muscoda – 12 miles to the west
  - Spring Green – 8 miles to the east



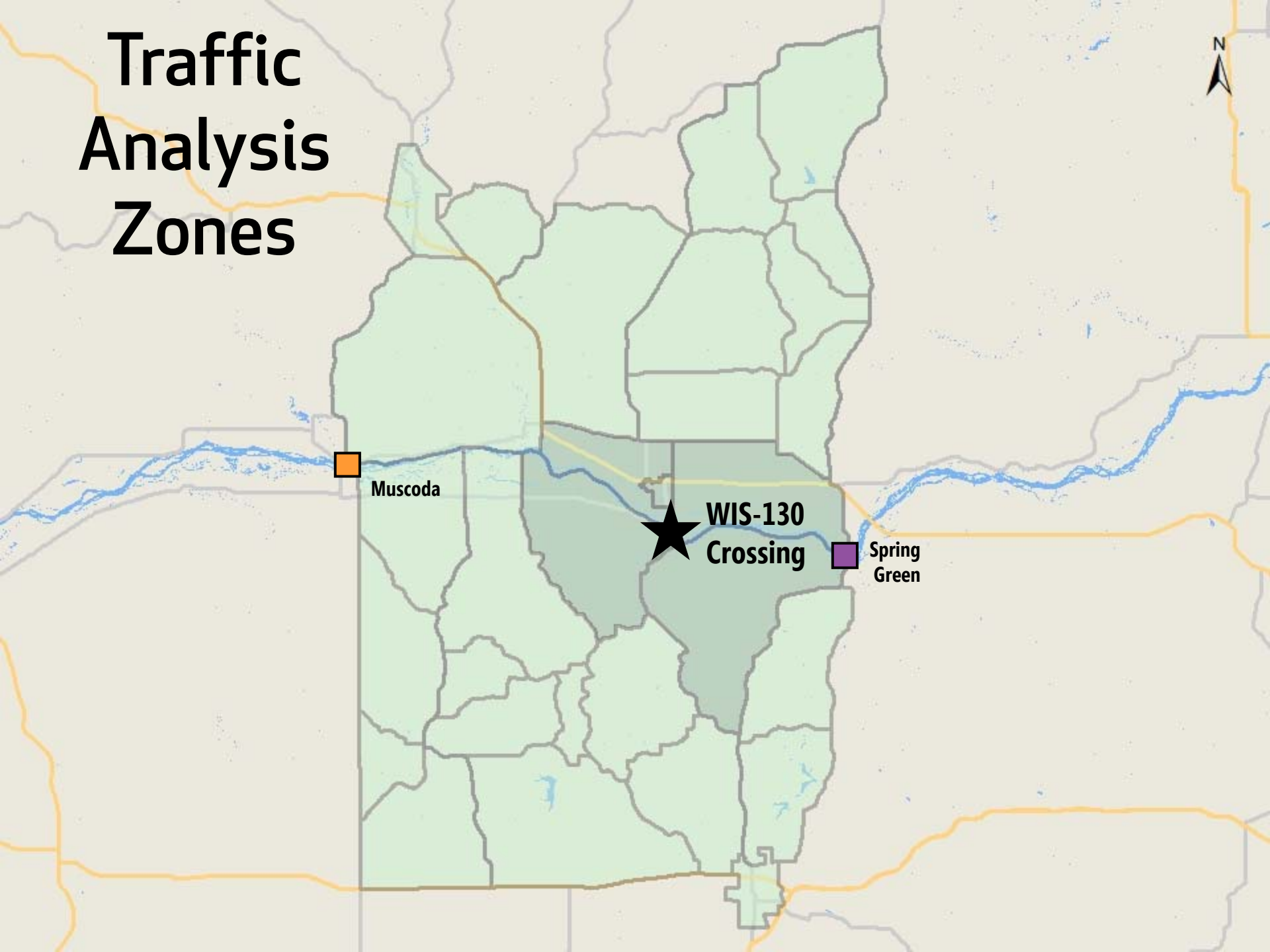
# Calibration

- Used AADT values available through WisDOT to calibrate the results
- Traffic volumes communicate the magnitude of the impact

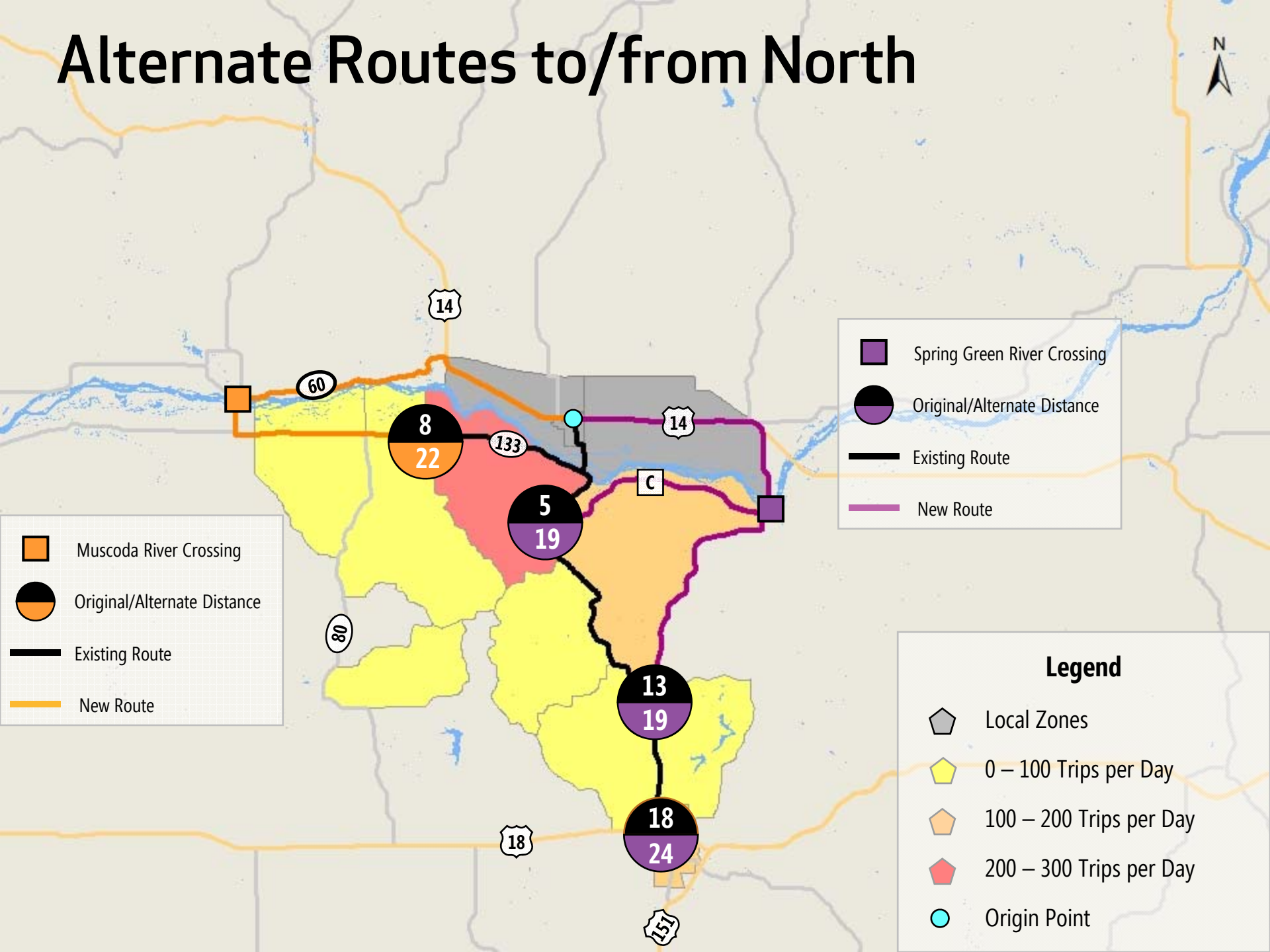




# Traffic Analysis Zones



# Alternate Routes to/from North



# Cost and Impact Summary

| <b>Alternative Number and Description</b> | <b>Alt. 1P<br/>Near Adjacent Structure<br/>Reconstruction (2 Bridges)</b> | <b>Alt. 5<br/>Planned Elimination</b> |
|-------------------------------------------|---------------------------------------------------------------------------|---------------------------------------|
| Centerline Length (Miles)                 | 0.7                                                                       | 0.7                                   |
| Project Cost Estimate (Million)           | \$40                                                                      | \$2-\$5                               |
| 20 Year Maintenance Costs (Million)       | < \$1                                                                     | N/A                                   |
| 21-75 Year Maintenance Costs (Million)    | \$11                                                                      | N/A                                   |
| Emergency Services Costs (Million)        | \$1-\$2 (during const)                                                    | \$2-\$4                               |
| 20 Year Road User Costs (Million)         | N/A                                                                       | \$27                                  |
| 21-75 Year Road User Costs (Million)      | N/A                                                                       | \$14                                  |
| Local Businesses Affected                 | N/A                                                                       | 20*                                   |
| Maintain Access to Long Island            | Yes                                                                       | No                                    |
| Maintain Access to Bakken Pond Woods      | No                                                                        | No                                    |
| Wetland Area (Net) (Acres)                | - <1                                                                      | + 4                                   |
| Total Area Converted to ROW (Acres)       | 10                                                                        | 0                                     |
| Total Costs to Society (Million)          | \$52 - \$53                                                               | \$45 - \$50                           |

# Questions?